Week 4 Day 4 — Model Tuning Report

Adult Census Income dataset | Search, diagnostics and final performance | Aqsa Saqib

1. Reproducible pipeline

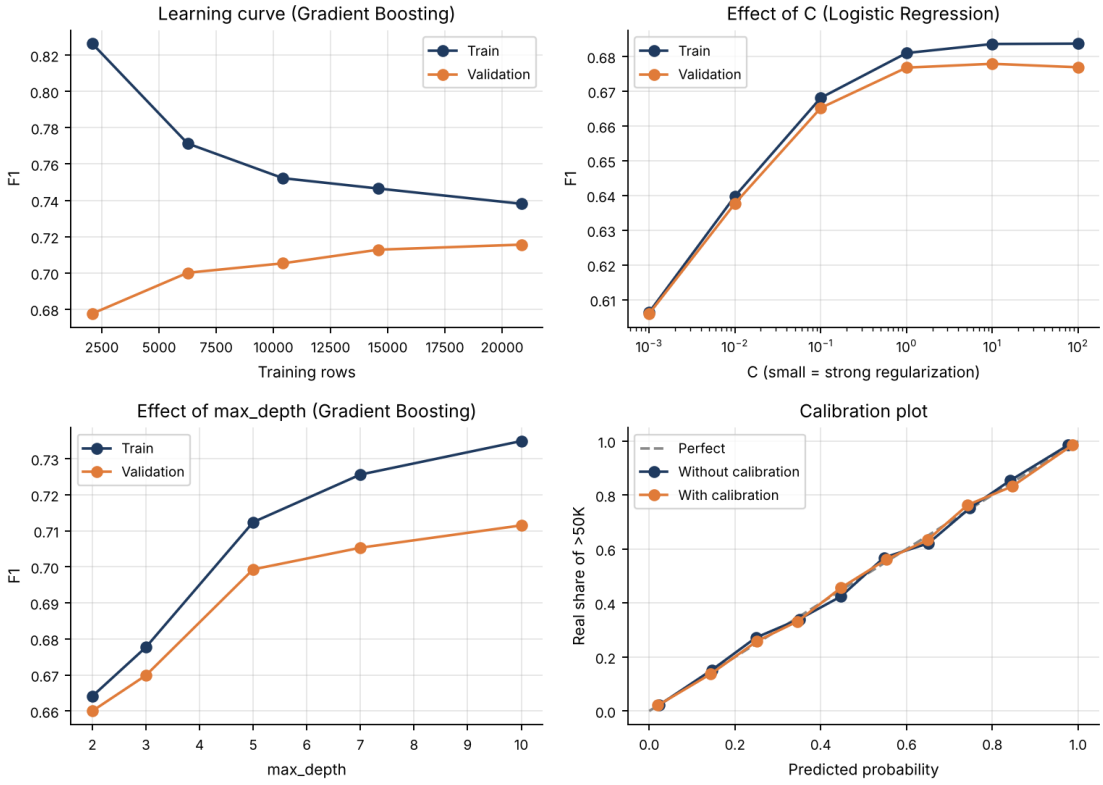
- **Pipeline:** make the 8 new features from Day 3 → preprocessing (median / "Missing", scaling, one-hot) → model.
- **Same results every time:** random_state = 42 everywhere and the same 5 folds for every step. The library versions are printed at the top of the notebook.
- **Test set:** not used until the final evaluation in section 5.

2. Hyperparameter search (5-fold CV, metric = F1)

Model	Search	What I searched	Best settings	F1 default → tuned
Logistic Regression	Grid search (8 settings)	penalty: l1, l2 C: 0.01, 0.1, 1, 10	penalty = l2 C = 10	0.676 → 0.678
Gradient Boosting	Randomized search (50 of 256 settings)	learning_rate: 0.03 - 0.2 max_iter: 100 - 500 max_depth: 3, 5, 7, None l2_regularization: 0 - 10	learning_rate = 0.03 max_iter = 300 max_depth = None l2_regularization = 10	**0.714 → 0.715**

- Tuning helped only a little. The default settings were already good.
- All top 5 Gradient Boosting settings use l2_regularization = 10, so holding the model back helps. Random Forest was third on Day 3, so I did not tune it.

3. Overfitting or underfitting?



- **Learning curve:** the gap between train and validation is small at the end (about 0.74 vs 0.72), so there is no strong overfitting. Validation is still going up a little, so **more data** would help a little.
- **C (Logistic Regression):** very small C underfits. From C = 1 the scores stop improving and train = validation. Stronger regularization will not help; a more flexible model helps more.
- **max_depth (Gradient Boosting):** depth 2 or 3 underfits. Deeper trees score better and the gap grows only a little, because l2_regularization = 10 holds them back.

4. Calibration & threshold

- **Calibration:** Brier score 0.0890 without calibration and 0.0892 with isotonic calibration. Both lines are on the diagonal in the calibration plot, so the probabilities are already good. **Decision: no calibration.**
- **Threshold:** chosen with out-of-fold predictions on the training data (not the test set).

Threshold	Precision	Recall	F1	People contacted
0.30	0.647	0.821	0.723	7,962
0.40 (chosen)	**0.710**	**0.747**	**0.728**	**6,598**
0.50 (default)	0.775	0.665	0.715	5,382
0.60	0.824	0.571	0.674	4,349

- **Threshold 0.5:** 5,382 people contacted, 4,169 real high earners reached, 1,213 wasted contacts.
- **Threshold 0.40:** 6,598 people contacted, 4,685 real high earners reached, 1,913 wasted contacts.
- So 0.40 reaches about 500 more real high earners but wastes about 700 more contacts. If wasted contacts are very expensive, 0.5 is the better choice.

5. Final evaluation on the hold-out test set

Final model = tuned Gradient Boosting, no calibration, threshold 0.40.

Metric	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC	Brier
Threshold 0.5	0.872	0.775	0.658	0.712	0.923	0.822	0.091
Threshold 0.40	**0.864**	**0.710**	**0.734**	**0.722**	**0.923**	**0.822**	**0.091**

- **Confusion matrix (threshold 0.40):** 4,474 true negatives, 471 false positives, 417 false negatives, 1,151 true positives.
- **Progress:** F1 0.505 (Day 1 baseline) → 0.667 (Day 2 Logistic Regression) → **0.722** (today).
- The test F1 (0.722) is close to the cross-validation F1 (0.728), so the model works well on new data.

6. Saved model & production note

The file **final_model.joblib** holds the full pipeline (features + preprocessing + model) and the threshold. Run the add_features cell first, then:

```
saved = joblib.load("final_model.joblib")
proba = saved["model"].predict_proba(new_people)[:, 1]
prediction = (proba >= saved["threshold"]).astype(int)   # 1 = earns >50K
```

- **Expected behaviour:** F1 around 0.72. About 7 in 10 people we contact will be real high earners, and we will find about 3 in 4 of all high earners.
- The data is from 1994, so the model must be checked and retrained on new data before real use. New data must have the same columns as adult.csv.